

Práctica Docker

Hecha en el wsl2 de Ubuntu, por Enrique Fernández Sanz

```
menique@DESKTOP-U2KFOO x + v
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Mar 15 18:42:48 CET 2026

System load: 0.0          Processes: 48
Usage of /:  0.5% of 1006.85GB Users logged in: 0
Memory usage: 2%          IPv4 address for eth0: 172.23.217.249
Swap usage: 0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

This message is shown once a day. To disable it please create the
/home/menique/.hushlogin file.
menique@DESKTOP-U2KFOO5:~$
```

```
menique@DESKTOP-U2KFOO5:~$ neofetch
.-/+00ssssso+/-.
':+ssssssssssssssssss+:`
--+ssssssssssssssssssysss+-
.ossssssssssssssssssdMMMMNyssso.
/sssssssssshdmmNNmmyNMMMMhssssss/
+ssssssssshmydMMMMMMNddddysssssss+
/sssssssshNMMMyhhyyyhmNMMMNhssssss/
.ssssssssdMMMMNhssssssssshNMMMdssssss+.
+ssssshhhyNMMNysssssssssyNMMMyssssss+
ossyNMMMNyMMhssssssssssshmmhssssssso
ossyNMMMNyMMhssssssssssshmmhssssssso
+ssssshhhyNMMNysssssssssyNMMMyssssss+
.ssssssssdMMMMNhssssssssshNMMMdssssss+.
/sssssssshNMMMyhhyyyhdNMMMNhssssss/
+ssssssssshmydMMMMMMNddddysssssss+
/sssssssshdmmNNmmyNMMMMhssssss/
.ossssssssssssssssssdMMMMNyssso.
--+ssssssssssssssssssysss+-
':+ssssssssssssssssss+:`
.-/+00ssssso+/-.

menique@DESKTOP-U2KFOO5
-----
OS: Ubuntu 24.04.4 LTS on Windows 10 x86_64
Kernel: 5.15.167.4-microsoft-standard-WSL2
Uptime: 8 mins
Packages: 770 (dpkg)
Shell: bash 5.2.21
Theme: Adwaita [GTK3]
Icons: Adwaita [GTK3]
Terminal: Relay(283)
CPU: AMD Ryzen 7 5800X (16) @ 3.800GHz
GPU: faaf:00:00:0 Microsoft Corporation Basic Render Driver
Memory: 422MiB / 15958MiB
```

Vamos primero a instalar docker (la práctica empieza en la diapositiva número 5).

\$ apt-get update

```
menique@DESKTOP-U2KF005:~$ apt-get update
Reading package lists... Done
E: Could not open lock file /var/lib/apt/lists/lock - open (13: Permission denied)
E: Unable to lock directory /var/lib/apt/lists/
W: Problem unlinking the file /var/cache/apt/pkgcache.bin - RemoveCaches (13: Permission denied)
W: Problem unlinking the file /var/cache/apt/srcpkgcache.bin - RemoveCaches (13: Permission denied)
menique@DESKTOP-U2KF005:~$ sudo !!
sudo apt-get update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
menique@DESKTOP-U2KF005:~$ |
```

\$ apt-get install apt-transport-https ca-certificates curl gnupg-agent software-properties-common

```
menique@DESKTOP-U2KF005:~$ sudo apt-get install apt-transport-https ca-certificates curl gnupg-agent software-properties-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.8).
curl set to manually installed.
software-properties-common is already the newest version (0.99.49.4).
software-properties-common set to manually installed.
The following NEW packages will be installed:
  apt-transport-https gnupg-agent
0 upgraded, 2 newly installed, 0 to remove and 27 not upgraded.
Need to get 8642 B of archives.
After this operation, 64.5 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-https all 2.8.3 [3970 B]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 gnupg-agent all 2.4.4-2ubuntu17.4 [4672 B]
Fetched 8642 B in 0s (27.3 kB/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 54220 files and directories currently installed.)
Preparing to unpack .../apt-transport-https_2.8.3_all.deb ...
Unpacking apt-transport-https (2.8.3) ...
Selecting previously unselected package gnupg-agent.
Preparing to unpack .../gnupg-agent_2.4.4-2ubuntu17.4_all.deb ...
Unpacking gnupg-agent (2.4.4-2ubuntu17.4) ...
Setting up apt-transport-https (2.8.3) ...
Setting up gnupg-agent (2.4.4-2ubuntu17.4) ...
menique@DESKTOP-U2KF005:~$ |
```

Vamos primero a instalar docker

(la práctica empieza en la diapositiva número 5).

```
$ sudo install -m 0755 -d /etc/apt/keyrings
$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
$ sudo chmod a+r /etc/apt/keyrings/docker.asc
```

```
menique@DESKTOP-U2KF005:~$ sudo install -m 0755 -d /etc/apt/keyrings
menique@DESKTOP-U2KF005:~$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
menique@DESKTOP-U2KF005:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
menique@DESKTOP-U2KF005:~$ |
```

```
$ tee /etc/apt/sources.list.d/docker.sources <<EOF
```

```
Types: deb
```

```
URIs: https://download.docker.com/linux/ubuntu
```

```
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
```

```
Components: stable
```

```
Signed-By: /etc/apt/keyrings/docker.asc
```

```
EOF
```

```
menique@DESKTOP-U2KF005:~$ sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Signed-By: /etc/apt/keyrings/docker.asc
EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: noble
Components: stable
Signed-By: /etc/apt/keyrings/docker.asc
menique@DESKTOP-U2KF005:~$ |
```

Vamos primero a instalar docker

(la práctica empieza en la diapositiva número 5).

\$ apt-get update

```
menique@DESKTOP-U2KF005:~$ sudo apt-get update
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:2 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [46.7 kB]
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 95.2 kB in 0s (221 kB/s)
Reading package lists... Done
menique@DESKTOP-U2KF005:~$ |
```

\$ apt-get install docker-ce docker-ce-cli containerd.io

```
menique@DESKTOP-U2KF005:~$ sudo apt-get install docker-ce docker-ce-cli containerd.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-buildx-plugin docker-ce-rootless-extras docker-compose-plugin iptables libip4tc2 libip6tc2 libnetfilter-contrack3 libnfnetlink0 libnftables1 libnftnl11
  libslirp0 nftables pigz slirp4netns
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin firewalld
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin iptables libip4tc2 libip6tc2 libnetfilter-contrack3
  libnfnetlink0 libnftables1 libnftnl11 libslirp0 nftables pigz slirp4netns
0 upgraded, 17 newly installed, 0 to remove and 27 not upgraded.
Need to get 98.2 MB of archives.
After this operation, 397 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
```

1. Demuestra la versión de Docker instalada y que el servicio está corriendo.

\$ docker version

```
menique@DESKTOP-U2KF005:~$ docker version
Client: Docker Engine - Community
 Version:           29.3.0
 API version:       1.54
 Go version:        go1.25.7
 Git commit:        5927d80
 Built:             Thu Mar  5 14:25:48 2026
 OS/Arch:           linux/amd64
 Context:           default

Server: Docker Engine - Community
 Engine:
  Version:           29.3.0
  API version:       1.54 (minimum version 1.40)
  Go version:        go1.25.7
  Git commit:        83bca51
  Built:             Thu Mar  5 14:25:48 2026
  OS/Arch:           linux/amd64
  Experimental:      false
 containerd:
  Version:           v2.2.2
  GitCommit:        301b2dac98f15c27117da5c8af12118a041a31d9
 runc:
  Version:           1.3.4
  GitCommit:        v1.3.4-0-gd6d73eb8
 docker-init:
  Version:           0.19.0
  GitCommit:        de40ad0
```

\$ systemctl status docker

```
menique@DESKTOP-U2KF005:~$ systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-03-15 19:09:23 CET; 5min ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 3121 (dockerd)
      Tasks: 16
     Memory: 46.2M (-)
    CGroup: /system.slice/docker.service
            └─3121 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Mar 15 19:09:23 DESKTOP-U2KF005 dockerd[3121]: time="2026-03-15T19:09:23.821154536+01:00" level=warning msg="WARNING: No blkio throttle.write_bps_device support"
Mar 15 19:09:23 DESKTOP-U2KF005 dockerd[3121]: time="2026-03-15T19:09:23.821161433+01:00" level=warning msg="WARNING: No blkio throttle.read_iops_device support"
Mar 15 19:09:23 DESKTOP-U2KF005 dockerd[3121]: time="2026-03-15T19:09:23.821164986+01:00" level=warning msg="WARNING: No blkio throttle.write_iops_device support"
Mar 15 19:09:23 DESKTOP-U2KF005 dockerd[3121]: time="2026-03-15T19:09:23.821168709+01:00" level=warning msg="WARNING: Support for cgroup v1 is deprecated and planned to be removed"
Mar 15 19:09:23 DESKTOP-U2KF005 dockerd[3121]: time="2026-03-15T19:09:23.821178800+01:00" level=info msg="Docker daemon" commit=83bca51 containerd-snapshotter=true storage=overlayfs
Mar 15 19:09:23 DESKTOP-U2KF005 dockerd[3121]: time="2026-03-15T19:09:23.821376772+01:00" level=info msg="Initializing buildkit"
Mar 15 19:09:23 DESKTOP-U2KF005 dockerd[3121]: time="2026-03-15T19:09:23.839042380+01:00" level=info msg="Completed buildkit initialization"
Mar 15 19:09:23 DESKTOP-U2KF005 dockerd[3121]: time="2026-03-15T19:09:23.842607588+01:00" level=info msg="Daemon has completed initialization"
Mar 15 19:09:23 DESKTOP-U2KF005 dockerd[3121]: time="2026-03-15T19:09:23.842685839+01:00" level=info msg="API listen on /run/docker.sock"
Mar 15 19:09:23 DESKTOP-U2KF005 systemd[1]: Started docker.service - Docker Application Container Engine.
menique@DESKTOP-U2KF005:~$
```

2. Descárgate las siguientes imágenes:

a. `alpine:latest`

a. `$ docker image pull alpine:latest`

b. `bitnami/git`

b. `$ docker image pull bitnami/git`

c. `Python:latest`

c. `$ docker image pull python:latest`

```
menique@DESKTOP-U2KF005:~$ docker image pull alpine:latest
latest: Pulling from library/alpine
failed to copy: httpReadSeeker: failed open: failed to do request: Get "https://docker-images-prod.6aa30f8b08e16409b46e0173d6de2f56.r2.cloudflarestorage.com/registry-v2/doc
ker/registry/v2/blobs/sha256/a4/a40c03cbb81c59bfb0e0887ab0b1859727075da7b9cc576a1cec2c771f38c5fb/data?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=f1baa2dd9b876aeb89ef
ebbf9e5d5f4%2F20260315%2Fauto%2Fs3%2Faws4_request&X-Amz-Date=20260315T181847Z&X-Amz-Expires=1200&X-Amz-SignedHeaders=host&X-Amz-Signature=6263b957ddb75aac3783c614ccc7209e9
c9f26c7ae9f19962fe338f3b11a4b75": dial tcp 172.64.66.1:443: i/o timeout
menique@DESKTOP-U2KF005:~$ docker image pull bitnami/git
Using default tag: latest
latest: Pulling from bitnami/git
57e614edf458: Pull complete
Digest: sha256:abd36369a4cc88d19b0a1eca521042afeff7f3142b56fd834f28697233cb7946
Status: Downloaded newer image for bitnami/git:latest
docker.io/bitnami/git:latest
menique@DESKTOP-U2KF005:~$ docker image pull python:latest
latest: Pulling from library/python
failed to copy: httpReadSeeker: failed open: failed to do request: Get "https://docker-images-pro
ker/registry/v2/blobs/sha256/d8/d8f278e9e42ab807052d815573564c27a66903eb8fdcfdd7ee128021b1db2660/
ebbf9e5d5f4%2F20260315%2Fauto%2Fs3%2Faws4_request&X-Amz-Date=20260315T182306Z&X-Amz-Expires=1200
f0ec52e24ae7136018f952655177f13": dial tcp 172.64.66.1:443: i/o timeout
menique@DESKTOP-U2KF005:~$ |
```

(He tenido que desactivar el firewall de windows, añadir la aplicación de Docker para que sea ignorado por el firewall, sincronizar el wsl2 en el Docker Desktop, configurar el daemon de Docker Desktop para que use un mirror de docker hub en vez de cloudflare R2 y reiniciar los drivers de ethernet de windows para hacer que se pudieran descargar las imagenes)

(Este es realmente un problema que solo le puede ocurrir a los 5 usuarios actuales que usan Windows para trabajar con contenedores en pleno 2026)

(He tardado 1 hora en resolver esto, ya que apenas he podido encontrar alguna solución en las redes)

```
menique@DESKTOP-U2KF005:~$ docker image pull alpine:latest
latest: Pulling from library/alpine
589002ba0eae: Pull complete
9e595aac14e0: Download complete
caa817ad3aea: Download complete
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
docker.io/library/alpine:latest
menique@DESKTOP-U2KF005:~$ docker image pull python:latest
latest: Pulling from library/python
6052218b0e44: Pull complete
a793e3c6bce8: Pull complete
36a6d998ca16: Pull complete
ab1b8c92e120: Pull complete
9da421ddeb65: Pull complete
866771c43bf5: Pull complete
ed881fbf1b07: Pull complete
028929fe143d: Download complete
8bbd577bbad9: Download complete
Digest: sha256:7aea6827c8787754f99339ffed8cfc41fb09421f9c7d0e77a198b08422a3455e
Status: Downloaded newer image for python:latest
docker.io/library/python:latest
menique@DESKTOP-U2KF005:~$ |
```

3. Muestra un listado de las imágenes descargadas.

\$ docker image ls

```
menique@DESKTOP-U2KF005:~$ docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	25109184c71b	13.1MB	3.95MB	
bitnami/git:latest	abd36369a4cc	1.26GB	347MB	
python:latest	7aea6827c878	1.63GB	432MB	

```
menique@DESKTOP-U2KF005:~$ |
```

 Info →  In Use

- a. Muestra el historial de una de ellas.

\$ docker image history [imagen]

```
menique@DESKTOP-U2KF005:~$ docker image history alpine
```

IMAGE	CREATED	CREATED BY	SIZE	COMMENT
25109184c71b	6 weeks ago	CMD ["/bin/sh"]	0B	buildkit.dockerfile.v0
<missing>	6 weeks ago	ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /...	9.11MB	buildkit.dockerfile.v0

```
menique@DESKTOP-U2KF005:~$ |
```

4. Ejecuta un contenedor de cada imagen de la siguiente manera:

a. `$ docker run <id_imagen>`

```
menique@DESKTOP-U2KF005:~$ docker image history alpine
IMAGE          CREATED          CREATED BY          SIZE      COMMENT
25109184c71b   6 weeks ago     CMD ["/bin/sh"]     0B        buildkit.dockerfile.v0
<missing>      6 weeks ago     ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /... 9.11MB    buildkit.dockerfile.v0
menique@DESKTOP-U2KF005:~$ docker run alpine
menique@DESKTOP-U2KF005:~$ docker run bitnami/git
 20:09:36.62 INFO ==>
 20:09:36.62 INFO ==> Welcome to the Bitnami git container
 20:09:36.62 INFO ==>
menique@DESKTOP-U2KF005:~$ docker run python
menique@DESKTOP-U2KF005:~$ |
```


5. Muestra el listado de todos los contenedores indicando:

- a. Su estado
- b. Su nombre
- c. Su id y la id de la imagen en la que se basan.

\$ docker ps -a

Si no le damos un nombre,
se genera

```
menique@DESKTOP-U2KF005:~$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
cabb493d1b7b	python	"python3"	2 minutes ago	Exited (0) 2 minutes ago		inspiring_feynman
6b1e90514f90	bitnami/git	"/opt/bitnami/script..."	2 minutes ago	Exited (0) 2 minutes ago		happy_babbage
73012c321d58	alpine	"/bin/sh"	2 minutes ago	Exited (0) 2 minutes ago		agitated_chatelet

```
menique@DESKTOP-U2KF005:~$ |
```

6. Intenta meterte dentro de un contenedor:

a. Porque no deja.

b. Cambia el estado al estado levantado "UP" e intenta meterte.

i. Indica porque no cambia de estado.

a. \$ docker attach [id contenedor]

```
CONTAINER ID   IMAGE          COMMAND
cabb493d1b7b   python        "python3"
6b1e90514f90   bitnami/git   "/opt/bitnami/script..."
73012c321d58   alpine        "/bin/sh"
menique@DESKTOP-U2KF005:~$ docker attach cabb493d1b7b
cannot attach to a stopped container, start it first
menique@DESKTOP-U2KF005:~$ |
```

b. \$ docker start [idcontenedor]
\$ docker attach [idcontenedor]

```
menique@DESKTOP-U2KF005:~$ docker start cabb493d1b7b
cabb493d1b7b
menique@DESKTOP-U2KF005:~$ docker attach cabb493d1b7b
cannot attach to a stopped container, start it first
```

i. El contenedor si no ha sido creado con -it, puede que no cambie a up

```
menique@DESKTOP-U2KF005:~$ docker stop python
Error response from daemon: No such container: python
menique@DESKTOP-U2KF005:~$ docker stop cabb493d1b7b
cabb493d1b7b
menique@DESKTOP-U2KF005:~$ docker run -it python
Python 3.14.3 (main, Feb 24 2026, 21:41:39) [GCC 14.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> print("Hello World");
Hello World
>>> |
```

```
menique@DESKTOP-U2KF005:~$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
51754c8a8364   python        "python3"               About a minute ago   Exited (0) 20 seconds ago         charming_haslett
cabb493d1b7b   python        "python3"               11 minutes ago     Exited (0) 4 minutes ago         inspiring_feynman
6b1e90514f90   bitnami/git   "/opt/bitnami/script..." 11 minutes ago     Exited (0) 11 minutes ago         happy_babbage
73012c321d58   alpine        "/bin/sh"               12 minutes ago     Exited (0) 12 minutes ago         agitated_chatelet
```

7. Borra todos los contenedores creados.

```
$ docker rm $(docker ps -a -f status=exited -q)
```

```
menique@DESKTOP-U2KF005:~$ docker ps -a -f status=exited -q
51754c0a8364
cabb493d1b7b
6b1e90514f90
73012c321d58
menique@DESKTOP-U2KF005:~$ docker rm $(!!)
docker rm $(docker ps -a -f status=exited -q)
51754c0a8364
cabb493d1b7b
6b1e90514f90
73012c321d58
menique@DESKTOP-U2KF005:~$ |
```

```
menique@DESKTOP-U2KF005:~$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED         STATUS         PORTS          NAMES
```

8. Ejecuta contenedores:

- a. Uno de cada imagen, y de la imagen alpine:latest lanza dos.
- b. Asígnales un nombre personalizado.
- c. Créalos de tal manera que posteriormente se puedan levantar.

```
$ docker run --name alpine1 -it alpine
```

```
$ docker run --name alpine2 -it alpine
```

```
$ docker run --name bitnami1 -it bitnami/git
```

```
$ docker run --name python1 -it python
```

```
menique@DESKTOP-U2KF005:~$ docker run --name alpine1 -it alpine
/ # exit
menique@DESKTOP-U2KF005:~$ docker run --name alpine2 -it alpine
/ # exit
menique@DESKTOP-U2KF005:~$ docker run --name bitnami1 -it bitnami
Unable to find image 'bitnami:latest' locally
docker: Error response from daemon: failed to resolve reference "docker.i
Run 'docker run --help' for more information
menique@DESKTOP-U2KF005:~$ docker run --name bitnami1 -it bitnami/git
20:30:52.67 INFO ==>
20:30:52.67 INFO ==> Welcome to the Bitnami git container
20:30:52.67 INFO ==>
root [ / ]# exit
exit
menique@DESKTOP-U2KF005:~$ docker run --name python1 -it python
Python 3.14.3 (main, Feb 24 2026, 21:41:39) [GCC 14.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> exit
menique@DESKTOP-U2KF005:~$ |
```

```
menique@DESKTOP-U2KF005:~$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS              PORTS          NAMES
adde94cae89   python    "python3"               30 seconds ago Exited (0) 26 seconds ago           python1
cb96a15c189c   bitnami/git "/opt/bitnami/script..." 53 seconds ago Exited (0) 50 seconds ago           bitnami1
79a2b0a9af7b   alpine    "/bin/sh"               About a minute ago Exited (0) About a minute ago           alpine2
9d22f6798d97   alpine    "/bin/sh"               About a minute ago Exited (0) About a minute ago           alpine1
menique@DESKTOP-U2KF005:~$ |
```

9. Cambia el estado de cada contenedor a levantado “UP”

- a. Abre cuatro terminales diferentes y métete dentro de los contenedores.
- En la máquina con la imagen de alpine:
 - Dentro de home crea un fichero con tu nombre.
 - En la máquina que contiene el git crea una carpeta dentro de “/home” llamada “miservidorgit” y dentro de la carpeta clona el siguiente repositorio “<https://github.com/carlosdatdaw/RepoTestDaw.git>”
- Voy a usar tmux, que me permite crear varias terminales y ventanas distintas.

- a. \$ docker start alpine1
- \$ docker start alpine2
- \$ docker start bitnami1
- \$ docker start python1

```
menique@DESKTOP-U2KF005:~$ docker start alpine1
alpine1
menique@DESKTOP-U2KF005:~$ |

menique@DESKTOP-U2KF005:~$ docker start alpine2
alpine2
menique@DESKTOP-U2KF005:~$

menique@DESKTOP-U2KF005:~$ docker start bitnami1
bitnami1
menique@DESKTOP-U2KF005:~$

menique@DESKTOP-U2KF005:~$ docker start python1
python1
menique@DESKTOP-U2KF005:~$

[0] 0: bash* "DESKTOP-U2KF005" 21:37 15-Mar-26
```

9. Cambia el estado de cada contenedor a levantado “UP”

a. Abre cuatro terminales diferentes y métete dentro de los contenedores.

i. En la máquina con la imagen de alpine:

1. Dentro de home crea un fichero con tu nombre.

ii. En la máquina que contiene el git crea una carpeta dentro de “/home” llamada “miservidorgit” y dentro de la carpeta clona el siguiente repositorio

“<https://github.com/carlosdatdaw/RepoTestDaw.git>”

Voy a usar tmux, que me permite crear varias terminales y ventanas distintas.

b. `$ docker attach alpine1`

`$ docker attach alpine2`

`$ docker attach bitnami1`

`$ docker attach python1`

```
menique@DESKTOP-U2KF005:~$ docker start alpine1
alpine1
menique@DESKTOP-U2KF005:~$ docker attach alpine1
/#

menique@DESKTOP-U2KF005:~$ docker start alpine2
alpine2
menique@DESKTOP-U2KF005:~$ docker attach alpine2
/#

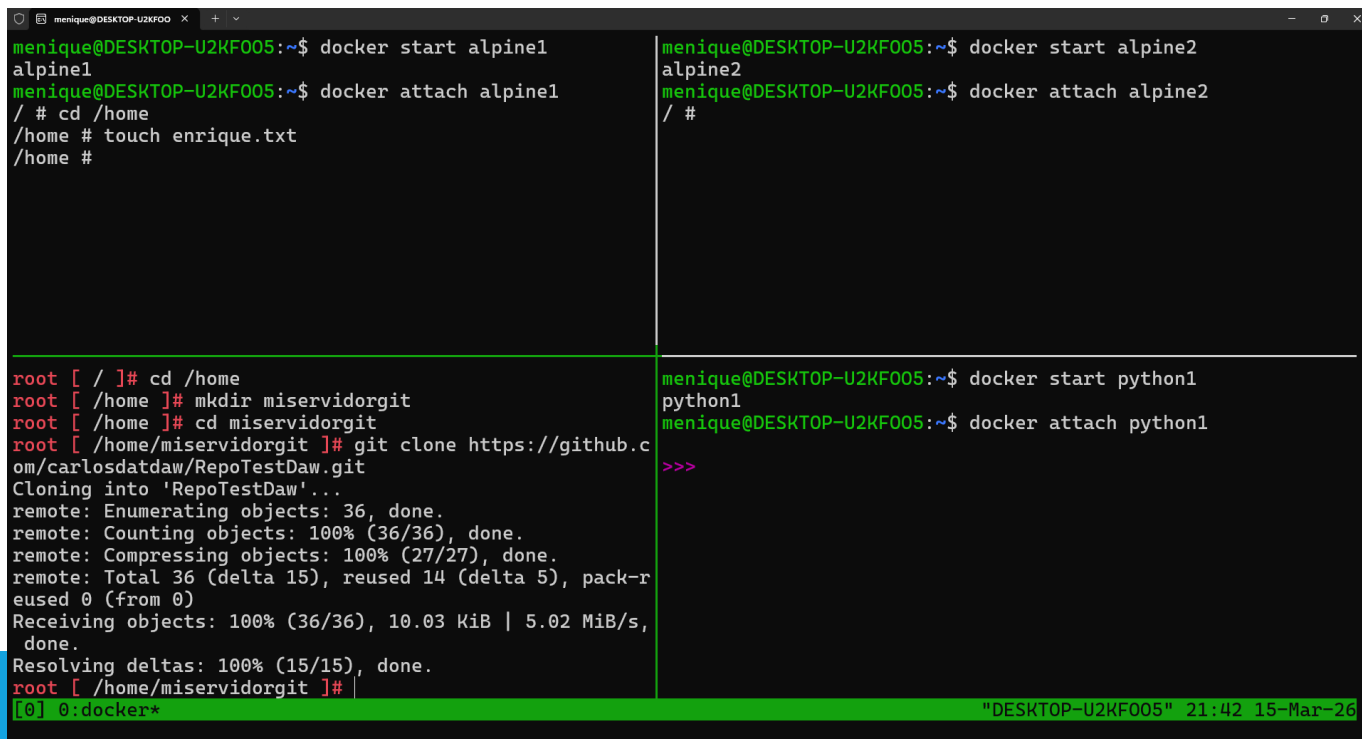
menique@DESKTOP-U2KF005:~$ docker start bitnami1
bitnami1
menique@DESKTOP-U2KF005:~$ docker attach bitnami1
root [ / ]#

menique@DESKTOP-U2KF005:~$ docker start python1
python1
menique@DESKTOP-U2KF005:~$ docker attach python1
>>> |

[0] 0:docker* "DESKTOP-U2KF005" 21:40 15-Mar-26
```

9. Cambia el estado de cada contenedor a levantado “UP”

- a. Abre cuatro terminales diferentes y métete dentro de los contenedores.
 - i. En la máquina con la imagen de alpine:
 - 1. Dentro de home crea un fichero con tu nombre.
 - ii. En la máquina que contiene el git crea una carpeta dentro de “/home” llamada “miservidorgit” y dentro de la carpeta clona el siguiente repositorio “<https://github.com/carlosdatdaw/RepoTestDaw.git>”



```
menique@DESKTOP-U2KF005:~$ docker start alpine1
alpine1
menique@DESKTOP-U2KF005:~$ docker attach alpine1
/ # cd /home
/home # touch enrique.txt
/home #

menique@DESKTOP-U2KF005:~$ docker start alpine2
alpine2
menique@DESKTOP-U2KF005:~$ docker attach alpine2
/ #

menique@DESKTOP-U2KF005:~$ docker start python1
python1
menique@DESKTOP-U2KF005:~$ docker attach python1
>>>

root [ / ]# cd /home
root [ /home ]# mkdir miservidorgit
root [ /home ]# cd miservidorgit
root [ /home/miservidorgit ]# git clone https://github.com/carlosdatdaw/RepoTestDaw.git
Cloning into 'RepoTestDaw'...
remote: Enumerating objects: 36, done.
remote: Counting objects: 100% (36/36), done.
remote: Compressing objects: 100% (27/27), done.
remote: Total 36 (delta 15), reused 14 (delta 5), pack-r
eused 0 (from 0)
Receiving objects: 100% (36/36), 10.03 KiB | 5.02 MiB/s,
done.
Resolving deltas: 100% (15/15), done.
root [ /home/miservidorgit ]#
```

[0] 0:docker* "DESKTOP-U2KF005" 21:42 15-Mar-26

9. Cambia el estado de cada contenedor a levantado “UP”

b. Desde la terminal inicial pon en estado de parado “STOP” una de las máquinas e indica que ha pasado.

Voy a crear otra terminal nueva. (La anterior era 0 Docker, esta sera 1 Bash)

b. `$ docker stop alpine2`

Se puede ver que al pararlo ya no puedo introducir comandos dentro de el.

```
menique@DESKTOP-U2KF005:~$ docker stop alpine2
alpine2
menique@DESKTOP-U2KF005:~$ |
```



```
menique@DESKTOP-U2KF005:~$ docker start alpine1
alpine1
menique@DESKTOP-U2KF005:~$ docker attach alpine1
/ # cd /home
/home # touch enrique.txt
/home #

menique@DESKTOP-U2KF005:~$ docker start alpine2
alpine2
menique@DESKTOP-U2KF005:~$ docker attach alpine2
/ # menique@DESKTOP-U2KF005:~$

menique@DESKTOP-U2KF005:~$ docker start python1
python1
menique@DESKTOP-U2KF005:~$ docker attach python1
>>>
```

root [/]# cd /home
root [/home]# mkdir miservidorgit
root [/home]# cd miservidorgit
root [/home/miservidorgit]# git clone https://github.com/carlosdatdaw/RepoTestDaw.git
Cloning into 'RepoTestDaw'...
remote: Enumerating objects: 36, done.
remote: Counting objects: 100% (36/36), done.
remote: Compressing objects: 100% (27/27), done.
remote: Total 36 (delta 15), reused 14 (delta 5), pack-r
eused 0 (from 0)
Receiving objects: 100% (36/36), 10.03 KiB | 5.02 MiB/s,
done.
Resolving deltas: 100% (15/15), done.
root [/home/miservidorgit]#

[0] 0:docker* 1:bash- "DESKTOP-U2KF005" 21:45 15-Mar-26

10. Crea una imagen a partir de los contenedores que se han creado datos anteriormente llámalos:

10. Crea una imagen a partir de los contenedores que se han creado datos anteriormente llámalos:

- imagengitpersonalv1
- imagenalpinepersonalv1
- Borra todos los contenedores que tengas.
- Crea un contenedor por cada una de las imágenes.
 - alpine:latest
 - bitnami:git
 - imagengitpersonalv1
 - imagenalpinepersonalv1

e. Comprueba que en el contenedor 1 y 2 no hay ningún tipo de datos creados, y que en el contenedor 3 y 4 si está la información creada anteriormente.

- `$ docker commit bitnami1 imagenbitnamipersonalv1`
- `$ docker commit alpine1 imagenalpinepersonalv1`
- `$ docker rm $(docker ps -a -f status=exited -q)`
- `$ docker run --name alpine_test -it alpine:latest`
`$ docker run --name git_test -it bitnami/git`
`$ docker run --name git_personal -it imagengitpersonalv1`
`$ docker run --name alpine_personal -it imagenalpinepersonalv1`

```
menique@DESKTOP-U2KF005:~$ docker start alpine_test
alpine_test
menique@DESKTOP-U2KF005:~$ docker attach alpine_test
/ # ls -lrt /home/
total 0
/ # exit
menique@DESKTOP-U2KF005:~$ docker start bitnami_test
bitnami_test
menique@DESKTOP-U2KF005:~$ docker attach bitnami_test
root [ / ]# ls -lrt /home/
total 0
root [ / ]# |
```

```
menique@DESKTOP-U2KF005:~$ docker commit bitnami1 imagenbitnamipersonalv1
sha256:44af7a3c6454073d497dad733279c53067f600cf38ea7e8230977de433221428
menique@DESKTOP-U2KF005:~$ docker commit alpine1 imagenalpinepersonalv1
sha256:ad97ce0134f6dd5041158f7c434266f35ef2a4cc8355fa62e07f5ec57e9f221d
```

```
menique@DESKTOP-U2KF005:~$ docker rm $(docker ps -a -f status=exited -q)
addde94cae89
cb96a15c189c
0d22f6798d97
menique@DESKTOP-U2KF005:~$ docker ps -a
```

```
menique@DESKTOP-U2KF005:~$ docker run --name alpine_test -it alpine:latest
/ # exit
menique@DESKTOP-U2KF005:~$ docker run --name bitnami_test -it bitnami/git
21:17:17.18 INFO ==>
21:17:17.18 INFO ==> Welcome to the Bitnami git container
21:17:17.19 INFO ==>
root [ / ]# exit
exit
menique@DESKTOP-U2KF005:~$ docker run --name bitnami_personal -it imagenbitnamipersonalv1
21:17:57.77 INFO ==>
21:17:57.77 INFO ==> Welcome to the Bitnami git container
21:17:57.77 INFO ==>
root [ / ]# exit
exit
menique@DESKTOP-U2KF005:~$ docker run --name alpine_personal -it imagenalpinepersonalv1
/ # exit
menique@DESKTOP-U2KF005:~$ |
```

```
menique@DESKTOP-U2KF005:~$ docker start alpine_personal
alpine_personal
menique@DESKTOP-U2KF005:~$ docker attach alpine_personal
/ # ls -lrt /home/
total 0
-rw-r--r-- 1 root root 0 Mar 15 20:41 enrique.txt
/ # exit
menique@DESKTOP-U2KF005:~$ docker start bitnami_personal
bitnami_personal
menique@DESKTOP-U2KF005:~$ docker attach bitnami_personal
root [ / ]# ls -lrt /home/
total 4
drwxr-xr-x 3 root root 4096 Mar 15 20:42 miservidorgit
root [ / ]# exit
exit
menique@DESKTOP-U2KF005:~$ |
```

11. Borra todos contenedores creados anteriormente.

```
$ docker rm $(docker ps -a -f status=exited -q)
```

```
menique@DESKTOP-U2KF005:~$ docker rm $(docker ps -a -f status=exited -q)
7e7d323de4a4
e67741775ef7
6455340eca22
2716a793432a
menique@DESKTOP-U2KF005:~$ |
```

11. Crea un volumen llamado “volum1alpine”:

a. Lanza un contenedor alpine:latest asociándole el volumen creado a la carpeta “/home” del contenedor.

b. Crea un fichero dentro de “/home” llamado “fichomevolum1.txt”

c. Borra el contenedor creado.

i. Verifica que dentro del volumen “volum1alpine” tiene el fichero “fichomevolum1.txt”

d. Crea de nuevo dos contenedores con la imagen alpine:latest asociándoles el volumen “volum1alpine” creado a la carpeta “/home” de los dos contenedores.

i. Verifica que en la carpeta “/home” de los dos contenedores existe el fichero “fichomevolum1.txt”

- a. `$ docker volume create volum1alpine`
`$ docker run --name alpinevol -it -v volum1alpine:/home alpine`
- b. `$ cd /home`
`$ touch fichomevolum1.txt`
- c. `$ docker rm alpinevol`
- c.i `$ docker volume inspect volum1alpine`
- d `$ docker run --name alpinevol2 -it -v volum1alpine:/home alpine`
`$ docker run --name alpinevol3 -it -v volum1alpine:/home alpine`
- d.i `$ ls /home`

```
menique@DESKTOP-U2KF005:~$ docker volume create volum1alpine
volum1alpine
```

```
menique@DESKTOP-U2KF005:~$ docker run --name alpinevol -it -v volum1alpine:/home alpine
/ # cd /home
/home # touch fichomevolum1.txt
/home # exit
menique@DESKTOP-U2KF005:~$ docker rm alpinevol
alpinevol
menique@DESKTOP-U2KF005:~$ docker volume inspect volum1alpine
[
  {
    "CreatedAt": "2026-03-15T21:39:38Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/volum1alpine/_data",
    "Name": "volum1alpine",
    "Options": null,
    "Scope": "local"
  }
]
menique@DESKTOP-U2KF005:~$ docker run --name alpinevol2 -it -v volum1alpine:/home alpine
/ # ls /home
fichomevolum1.txt
/ # exit
menique@DESKTOP-U2KF005:~$ docker run --name alpinevol3 -it -v volum1alpine:/home alpine
/ # ls /home
fichomevolum1.txt
/ # exit
menique@DESKTOP-U2KF005:~$ |
```

13. Borra todos contenedores creados anteriormente.

```
$ docker rm $(docker ps -a -f status=exited -q)
```

```
menique@DESKTOP-U2KF005:~$ docker rm $(docker ps -a -f status=exited -q)  
c0527dda4ec6  
aa1bba7ccf85  
menique@DESKTOP-U2KF005:~$ |
```

14. Descarga la imagen nginx:alpine y crea un contenedor con la siguiente información:

- a. Nombre: miservidor1
- b. Puerto 8080:80
- c. Lánzalo en segundo plano:
 - i. Mira el estado del contenedor.
 - ii. Mira en firefox si ya se ha levantado el servidor.

```
$ docker image pull nginx:alpine
```

```
$ docker run --name miservidor1 -p 8081:80 -d nginx:alpine  
(por alguna razón el 8080:80 esta ocupado, usare 8081:80)
```

```
$ docker ps
```

```
menique@DESKTOP-U2KF005:~$ docker image pull nginx:alpine
alpine: Pulling from library/nginx
c24026275c33: Pull complete
e19aff8f2cce: Pull complete
1549d7aec962: Pull complete
d2a46166eee6: Pull complete
593488f95c35: Pull complete
1f25242adbdb: Pull complete
c32126d2b96c: Pull complete
aa7a95392619: Download complete
0571eaf54e9d: Download complete
Digest: sha256:f46cb72c7df02710e693e863a983ac42f6a9579058a59a35f1ae36c9958e4ce6
Status: Downloaded newer image for nginx:alpine
docker.io/library/nginx:alpine
```

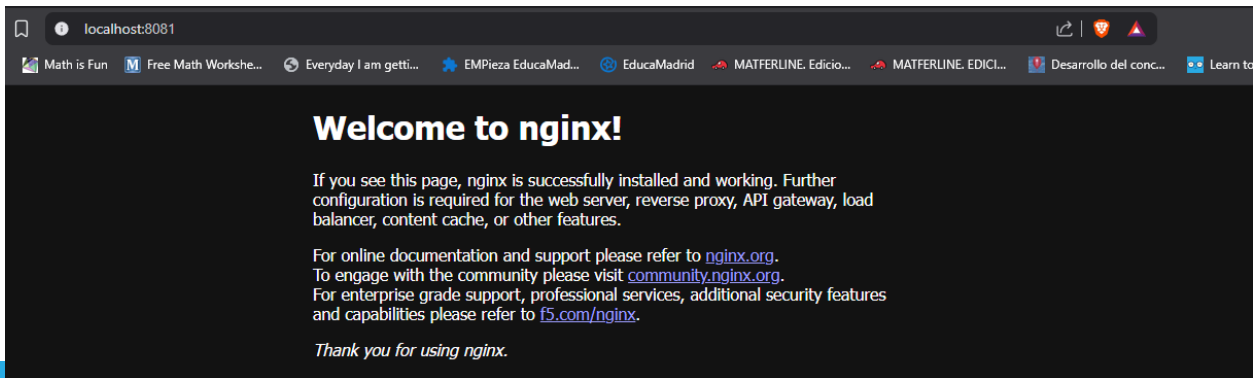
```
menique@DESKTOP-U2KF005:~$ docker run --name miservidor1 -p 8081:80 -d nginx:alpine
```

```
bc3e2a223fea4a3dcca3b4147701b61fa13d0875f1d00fd0dd231f51066a3f95
```

```
menique@DESKTOP-U2KF005:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
bc3e2a223fea	nginx:alpine	"/docker-entrypoint..."	2 minutes ago	Up 2 minutes	0.0.0.0:8081->80/tcp, [::]:8081->80/tcp	miservidor1

```
menique@DESKTOP-U2KF005:~$
```



```
menique@DESKTOP-U2KF005:~$ docker run --name miservidor1 -p 8080:80 -d nginx:alpine
c993ad2fc428c66edf5f88cf3075771faa73b04cb5340388e41e2b3872098568
docker: Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:8080 -> 127.0.0.1:0: /forwards/expose returned unexpected status: 500
```

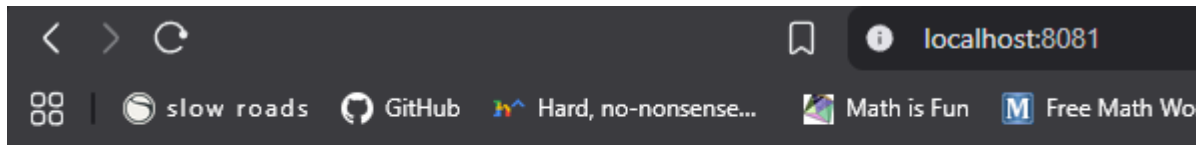
15. Descarga la imagen nginx:alpine y crea un contenedor con la siguiente información:

- a. Nombre: miservidor1
- b. Puerto 8081:80
- c. Crea un volumen llamado "mivolserv2" y guarda un fichero "index.html" con un ejemplo de una página web.
- d. Asóciate un volumen llamado "mivolserv2" a la carpeta del contenedor /usr/share/nginx/html
- e. Lánzalo en segundo plano:
 - i. Mira el estado del contenedor.
 - ii. Mira en firefox si ya se ha levantado el servidor y aparece la nueva página.

```
$ docker volume create mivolserv2
$ echo "<h1>Mi servidor web docker</h1>" > index.html
$ docker run --name miservidor2 -p 8081:80 -v mivolserv2:/usr/share/nginx/html -d
  nginx:alpine
$ docker cp index.html miservidor2:/usr/share/nginx/html/index.html
```

```
menique@DESKTOP-U2KF005:~$ docker volume create mivolserv2
mivolserv2
menique@DESKTOP-U2KF005:~$ echo "<h1>Mi servidor web docker</h1>" > index.html
```

```
menique@DESKTOP-U2KF005:~$ docker run --name miservidor2 -p 8081:80 -v mivolserv2:/usr/share/nginx/html -d nginx:alpine
5fefdf990ce3439d868835b76c57bdf11240807399601be22ca98f905e95608
menique@DESKTOP-U2KF005:~$ docker cp index.html miservidor2:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to miservidor2:/usr/share/nginx/html/index.html
menique@DESKTOP-U2KF005:~$ |
```



Mi servidor web docker

16. Descarga la imagen nginx:alpine y crea un contenedor con la siguiente información:

- Nombre: miservidor1
- Puerto 8082:80
- Crea una carpeta llamada "mipaginaweb" dentro del "/home" del usuario actual y guarda un fichero "index.html" con un ejemplo de una página web.
- Asóciate la carpeta "mipaginaweb" a la carpeta del contenedor /usr/share/nginx/html
- Lánzalo en segundo plano:
 - Mira en firefox si ya se ha levantado el servidor y aparece la nueva página.

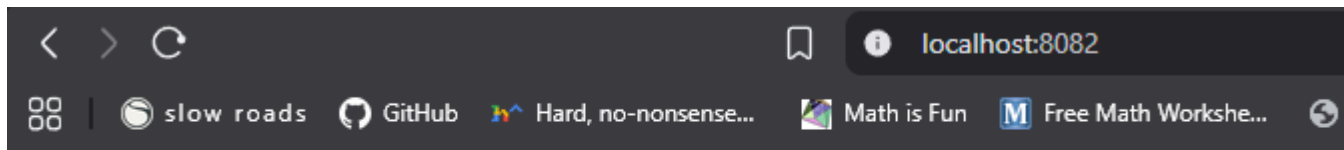
c. `$ mkdir /home/$USER/mipaginaweb`

`$ echo "<h1>Mi página desde el host</h1>" > /home/$USER/mipaginaweb/index.html`

`$ docker run --name miservidor3 -p 8082:80 -v /home/$USER/mipaginaweb:/usr/share/nginx/html -d nginx:alpine`

```
menique@DESKTOP-U2KF005:~$ mkdir mipaginaweb
menique@DESKTOP-U2KF005:~$ echo "<h1>Mi página desde el host</h1>" > ~/mipaginaweb/index.html
```

```
menique@DESKTOP-U2KF005:~$ docker run --name miservidor3 -p 8082:80 -v /home/menique/mipaginaweb:/usr/share/nginx/html -d nginx:alpine
3cf40003ce4f1eeacbbeaa64aa79aa01b71be2387191f0a2eed08f6ef746c2df
```



Mi pÃ¡gina desde el host